Pratyush Das

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Education

Purdue University August, 2021 - May, 2027 (Expected)

PhD in Computer Science (Advisors - Milind Kulkarni, Ben Delaware)

Research interests - Compilers, Automatic parallelization, Sparse tensors, High performance computing

Institute of Engineering & Management, Kolkata (MAKAUT) August, 2017 - May, 2021

Bachelor of Technology in Computer Science and Engineering

Awarded the Director's Award for Best Scientific Mind

Experience

ML Performance (Google Cloud) - Student Researcher May, 2026 - August, 2026

Google

- XLA TPU compiler: Built out an XLA plugin estimating TPU cycles of HLO ops without actually running on TPUs
  - Broadened HLO op coverage (e.g. all-reduce) and added support for more LLO emitters (pipeline, collectives).
  - Reduced delta against measured TPU runs on Gemini and YouTube workloads from over 50% to ∼5%.
  - Sped up static estimation by implementing compiler passes such as loop fast-forwarding.

SwiftUI - Compiler Intern May, 2023 - August, 2023

Apple

- Swift compiler: Extended the compiler and runtime to retarget SwiftUI to a new IR
  - Extended Swift's code generation and runtime, with LLVM patches, to lower SwiftUI to a new target IR.
  - Integrated an interpreter into Swift's runtime, enabling live updates of SwiftUI views from newly loaded code.
  - Implemented dynamic loading and symbol resolution of system shared libraries for interpreted code.

Google Summer of Code - Student June, 2021 - August, 2021

The LLVM Compiler Infrastructure Organization

Supervisors - William Moses, Johannes Doerfert

- Enzyme: LLVM Pass to perform automatic differentiation of statically analyzable LLVM IR
  - Integrated custom derivatives of several BLAS functions into Enzyme.
  - Wrote an LLVM pass to inline function definitions from bitcode files into LLVM IR.

IRIS-HEP - Fellow June, 2020 - September, 2020

Princeton University

Supervisor - Jim Pivarski

- Awkward Array: Library for nested, variable-sized data using NumPy-like idioms
  - Implemented a source-to-source compiler from a C++ subset to equivalent Python, with property-based testing.
  - Created a source-to-source compiler to generate equivalent parallel CUDA from specification (Python and type info).
  - Enabled automatic parallelization of user-written Awkward Array kernels.

IRIS-HEP and DIANA-HEP - Fellow June, 2018 - September, 2018; June, 2019 - September, 2019

Fermi National Accelerator Laboratory and Princeton University

Supervisor - Jim Pivarski

- Uproot: Python implementation of ROOT I/O, an open source file format storing over an exabyte of HEP data
  - Enabled writing fundamental HEP data structures like TTrees and histograms to ROOT files.
  - Uproot has become one of the most widely used HEP libraries.

Publications

- **P.Das**, A.Basareh, A.Pelenitsyn, K.Sundararajah, M.Kulkarni, B.Delaware, "Bring Your Own Formats and Kernels: Composable Abstractions for Sparse Matrix Computation", IEEE/ACM International Symposium on Code Generation and Optimization (CGO, 2027).
- **P.Das**, A.Xhebraj, T.Rompf, "Specializing Data Access in a Distributed File System (Generative Pearl)", 23rd ACM SIGPLAN International Conference on Generative Programming: Concepts and Experiences (GPCE, 2024), DOI: https://doi.org/10.1145/3689484.3690736.
- T.Mustafa, K.Kallas, **P.Das**, N.Vasilakis, "DiSh: Dynamic Shell-Script Distribution", 20th USENIX Symposium on Networked Systems Design and Implementation (NSDI, 2023).
- J.Pivarski, I.Osborne, **P.Das**, D.Lange, P.Elmer, "AwkwardForth: accelerating Uproot with an internal DSL", 25th International Conference on Computing in High-Energy and Nuclear Physics (vCHEP, 2021), DOI: 10.1051/epjconf/202125103002.

- J.Pivarski, I.Osborne, **P.Das**, A.Biswas, P.Elmer, "Awkward Array: JSON-like data, NumPy-like idioms", Proceedings of the 19th Python in Science Conference (SciPy USA, 2020), Pages 68-74, DOI: 10.25080/Majora-342d178e-00b.
- E.Rodrigues, et al., "The Scikit HEP Project - overview and prospects", Proceedings of the 24th International Conference on Computing in High Energy and Nuclear Physics (CHEP, 2019), DOI: 10.1051/epjconf/202024506028.
- N.Saha, **P.Das**, H.N.Saha, "Authorship Attribution of Short Texts using a Multi Layer Perceptron", International Journal of Applied Pattern Recognition, 2018 Vol. 5 No. 3, Pages 251-259, DOI: 10.1504/IJAPR.2018.10016100.

Other Open Source Contributions

Supervisor - Vassil Vassilev — November, 2019 - May, 2021

- **ROOT**: An open-source data analysis framework storing over an exabyte of data
  - Improvements to interpreter (rootcling)
- **Cling**: Interactive C++ interpreter built on top of Clang
  - Maintained cpt.py installer and packager
- **Clang**: C language family frontend for LLVM
  - Several patches to print type information of C++ template arguments

Supervisors - Jim Pivarski, Viktor Khristenko — June, 2017 - August, 2017

- **spark-root** - Apache Spark datasource for ROOT
  - Separated spark bindings from TTree reading code
- **root4j** - Java implementation of ROOT file reader
  - Optimized codebase to facilitate interoperability

Teaching Experience (Teaching Assistant)

CS 307: Software Engineering - Purdue University — Spring 2025, Fall 2025
CS 354: Operating Systems - Purdue University — Fall 2022 - Fall 2024, Spring 2026
CS 240: Programming in C - Purdue University — Fall 2021, Spring 2022

Programming Languages and Tools

Experienced: Python, CUDA, XLA
Familiar: C, C++, Java, Scala, Coq, Bash, LaTeX, Swift, WebAssembly, MLIR

Summer Schools

Oregon Programming Languages Summer School - University of Oregon — 2021
Computational and Data Science for High Energy Physics - Princeton University — 2019

Talks at Conferences

- Specializing Data Access in a Distributed File System (GPCE) — 2024
- GSoC Experience - Enzyme (LLVM Developers' Meeting) — 2021
- Python in High Energy Physics (SciPy India, PyCon USA) — 2019, 2020
- Writing files with uproot (PyHEP) — 2019
- Writing files with uproot (ROOT Users' Workshop) — 2018

Invited talks at Research Groups

- Staging Blocked Evaluation of Sparse Matrix Computations (PurPL Seminar) — 2026
- Language Transformations for the Awkward Array library (IRIS-HEP Fellow Presentations) — 2020
- CUDA backend for the Awkward Array project (Princeton University Liberty Research Group) — 2020
- Writing TTrees with uproot (IRIS-HEP Topical Meeting: Summer student project presentations) — 2019
- Writing files with uproot (DIANA Meeting: Updates on ROOT I/O) — 2018
- Separation of Concerns - Refactoring code between ROOT4J and Spark-Root (CMS Big Data Science, DIANA-HEP) — 2017

Posters

- SABLE: Staging Blocked Evaluation of Sparse Matrix Computations (Midwest PL Summit) — 2024

Service

- Purdue Programming Languages group (PurPL) co-ordinator — 2024,2025
- Artifact Evaluation Committee (SOSP) — 2023